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Immunosenescence and cancer: mechanisms and therapeutic targets

ABSTRACT

Immunosenescence contributes to tumor progression and shapes the tumor microenvironment.
Keywords: Immunosenescence; Cancer; Aging

Concept, process, and hallmarks of immunosenescence

Immunosenescence is a multifactorial process driven by intrinsic and extrinsic factors across the lifespan.

T cell aging

Aged T cells show impaired homeostasis. Therefore, degeneration of the thymus limits output and diversity.

Table 1 Characteristics of immunosenescence cells

Category	Markers	Cell types
Surface marker	CD27, CD28	T cells
Cell cycle	P16, P21	All
Cytokines	IL-6, IL-8	T cells

These markers define distinct differentiation states among leukocyte lineages during aging.

Conclusion

Immunosenescence is a key driver of tumor progression.

References

2 Hsu T. Educational initiatives in geriatric oncology. 2020.
3 Munoz-Espin D, Serrano M. Cellular senescence. 2019.
10 Pawelec G. Age and immunity. 2021.